



Reading a drive's SMART results

A drive keeps its own health records. SMART — Self-Monitoring, Analysis and Reporting Technology — is the part of a drive's firmware that counts its errors, tracks its age and temperature, and runs self-tests on request. The operating system does not grade the drive; the drive grades itself, and `smartctl` is how you ask to see the marking.

This unit covers reading those records: the one command that shows everything, the five numbers that decide whether a drive can be trusted, and the two traps that make healthy drives look dying and dying drives look healthy.



NOTE

A self-test runs inside the drive, not on the computer. The terminal that started it will never print a result, and closing that terminal changes nothing. The result waits in the drive's own log until something asks for it.

The command

```
sudo smartctl -H -A -l selftest /dev/sda
```

Part	What it does
<code>sudo</code>	Drive-level queries need administrator rights.
<code>smartctl</code>	The control tool from the <code>smartmontools</code> package.
<code>-H</code>	The drive's overall verdict on itself — one line, <code>PASSED</code> or <code>FAILED</code> .
<code>-A</code>	The attribute table: the drive's running counters.
<code>-l selftest</code>	The self-test log, including the test you started.
<code>/dev/sda</code>	Which drive to ask.



TIP

Device names are handed out at boot and can swap between boots: the drive called `sda` today can be `sdb` tomorrow. Before acting on any result, confirm you are reading the drive you think you are: `sudo smartctl -i /dev/sda` prints the model and serial number, which match the label on the physical drive.



Prefer a window? Ubuntu ships **Disks** (search the applications grid for "Disks"). Select the drive, open the menu in the title bar, and choose **SMART Data & Self-Tests**. Same numbers, same log, and it can start and stop tests too.

Three layers, three different weights

`smartctl` answers at three levels, and they are not equally trustworthy.

1. **The overall verdict (-H)** is the weakest. `PASSED` means only that no counter has yet crossed its failure threshold — drives routinely report `PASSED` while visibly deteriorating. Treat it as "not already condemned", never as "healthy".
2. **The self-test log (-l selftest)** is the exam result. A long test reads every sector on the platter, so "completed without error" means the whole surface has actually been read, recently.
3. **The attribute table (-A)** is the continuous record. It is where deterioration shows first, and the *trend* across weeks matters more than any single reading.

Reading the self-test log

The log lists recent tests, newest first:

Num	Test	Status	Remaining	LifeTime(hours)
# 1	Extended	Completed without error	00%	10831
# 2	Short	Completed without error	00%	9905

Status	Meaning	What to do
Completed without error	Every sector read cleanly.	The pass you wanted.
Completed: read failure	Bad media found; the log records where.	Treat the drive as failing — see the verdicts below.
Self-test routine in progress	Still running; the percentage counts <i>down</i> to 00%.	Wait. Nothing else.
Interrupted (host reset)	A reboot or suspend cut it short.	No verdict either way. Run it again.
Aborted by host	Something cancelled it.	Run it again.



CHECK

`LifeTime(hours)` is the drive's power-on-hours count at the moment that test ran. Compare it against `power_On_Hours` in the attribute table: if the two match to within a few hours, the top entry is the test you just ran, not a relic from years ago.

The five numbers that decide



Backblaze, which operates hundreds of thousands of drives and publishes its failure statistics, tracks five SMART attributes as failure predictors: in its data, **76.7% of failed drives showed a non-zero value in at least one of these five** before failing. Read the **RAW_VALUE** column for all of them.

ID	Attribute	What it counts
5	Reallocated_Sector_Ct	Sectors that failed and were quietly replaced from the drive's spare pool.
187	Reported_Uncorrect	Errors the drive had to report back to the computer as uncorrectable.
188	Command_Timeout	Operations that took so long the drive gave up on them.
197	Current_Pending_Sector	Sectors failing to read <i>right now</i> , awaiting a verdict.
198	Offline_Uncorrectable	Sectors that could not be read even during the drive's own scans.

A healthy drive shows **zero** for all five. The lifecycle behind the two sector counts: a sector that fails a read goes *pending* (197). If a later read succeeds, it leaves the list; if the spot is confirmed bad, the drive swaps in a spare and the count moves to *reallocated* (5). Reallocation is the drive consuming its finite stock of spares — the mechanism working as designed, but the need for it is the warning.

WARNING



The Seagate panic trap. On Seagate drives, `Raw_Read_Error_Rate` (1), `Seek_Error_Rate` (7) and `Hardware_ECC_Recovered` (195) show raw values in the millions or billions. These are not errors: Seagate packs the total count of operations into the low bits of the same raw number, so it grows enormous on a perfectly healthy drive while the actual error count — the high bits — stays at zero. Judge those three by their normalised `VALUE` holding steady against `THRESH`, or ignore them. Only the five attributes above are read raw.

TIP



`UDMA_CRC_Error_Count` (199) counts data corrupted on the way between the drive and the motherboard. A rising 199 with the five above at zero is a **cable** fault, not a drive fault: reseal or replace the SATA data cable. The count never resets, so what matters is whether it stops rising.

The other columns, briefly

Each attribute row also carries `VALUE`, `WORST` and `THRESH` — a normalised score that starts high (typically 100 or 200), falls as the drive degrades, and fails when it meets `THRESH`; `WHEN_FAILED` says whether that has ever happened. These columns are what the `-H` verdict is built from, which is exactly why `-H` is slow to alarm: for the five counters above, the raw value moves long before the normalised score follows.

Two attributes are context rather than verdicts. `Power_On_Hours` (9) is the drive's total working life, useful for judging how much a clean record actually covers — a drive that has spent most of its years switched off has had few chances to discover its own bad sectors, which is precisely what a

long test corrects. `Temperature_Celsius` (194) should sit comfortably below about 40 °C in steady use.

What the results decide

Result	Verdict
Long test completed without error, all five counters at zero	Fit for its role. The surface has actually been read, end to end, recently.
Test passes, but a counter is small and non-zero	Usable, not trustworthy: nothing irreplaceable lives on it. Re-test in a few weeks — growth is the signal, not the absolute number.
Read failure, or counters climbing between checks	Plan its retirement. Copy anything wanted off it now, while it still reads, starting with what matters most.



NOTE

One clean pass is evidence, not immunity. Backblaze's same figure read the other way: roughly a quarter of its failed drives showed no warning in any of the five attributes. A backup schedule, not a test result, is what protects data — the test says whether a drive is fit to hold data today; the backup is for the drive that fails without notice tomorrow.

Keeping watch without doing this by hand

The `smartmontools` package includes **smartd**, a background service that polls every drive on a schedule, runs periodic self-tests, and raises an alert when a counter moves. Its configuration lives in `/etc/smartd.conf`. Setting it up belongs with monitoring and alerting, in a later unit — the point here is that the manual check this unit teaches is also available as a standing guard.

Re-run a long test before trusting a drive with a new role, after the machine is physically moved, and on something like a yearly rhythm in between. The **Disks** application shows a one-line assessment for every drive whenever the quick answer is enough.

Sources: Backblaze, "What SMART Stats Tell Us About Hard Drives" (backblaze.com/blog/hard-drive-smart-stats), which names the five attributes and the 76.7% figure; `smartmontools` documentation (`smartctl(8)`, smartmontools.org) for flag behaviour and self-test statuses; the Seagate raw-value encoding is community-documented (`smartmontools` issue tracker, TrueNAS forums) rather than published by Seagate, and is stated here with that weight. Checked 2026-08-09.

